

PROJECT STUDY GUIDE

# Zero-to-100 Project Knowledge

macOS-Inspired Desktop Portfolio

Aditya Goswami

HTML5

CSS3

Vanilla JS

29 Sections

## START HERE

# Top 10 Must-Remember Points

If you only revise one page before an interview, read this one.

1

## What it is

A macOS-style desktop **portfolio** built with plain **HTML + CSS + Vanilla JS**. Zero dependencies, no backend.

2

## Windows

Each folder shortcut opens a **draggable window**. JS adds the class `window-open`; CSS makes it visible.

3

## Specificity bug

Hide rule scored **0,3,1** and beat `.window-open` (**0,2,1**). Fix: `section:nth-of-type(n+3) = 0,1,1`.

4

## Event delegation

ONE click listener on `#desktop`, using `event.target.closest('[data-window]')`.

5

## Z-index stacking

`highestZIndex` starts at 100 and goes up by 1 on every focus. Newest window = on top.

6

## Dock vs shortcuts

Shortcuts are `<button>` and work. Dock icons are `<div>` and are decoration only.

7

## Contact form

Does NOT send email. `preventDefault()` cancels the submit. Needs Formspree / EmailJS.

8

## AI Assistant

Not real AI. Just an "In Progress" placeholder, two paragraphs.

9

## Git status

Not initialized, no `.git` folder. `index.html` and `01_mp.html` are identical.

10

## Still to do

Real links, resume PDF, git init, GitHub Pages deploy, compress images, delete orphaned `wallpaper.jpg`.

## NAVIGATE

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**How to read this guide:** Highlighted text = key term • pink chips = code • ✓ done ✗ not done ! warning

# 01 What This Project Is

This is a **personal portfolio website** for Aditya Goswami, an aspiring Full-Stack Developer.

**The concept:** Instead of a normal vertical-scroll webpage, the portfolio looks and behaves like a **macOS (Apple Mac) desktop**. When you visit the website you see:

- A dark wallpaper (glowing planet/sphere)
- A top menu bar (like a Mac's menu bar) with the Apple logo, app title, date, and clock
- Folder icons on the left side of the "desktop" (like real Mac desktop folder shortcuts)
- A Dock at the bottom (like the Mac app dock) with app icons

Each folder shortcut opens a "window" — which is a portfolio section (About Me, Projects, Skills, etc.) that appears as a floating macOS-style application window with a title bar, close/minimize/maximize buttons, and the ability to be dragged around the screen.

**What it is NOT:**

- It is not a real operating system
- It is not a React/Vue/Angular app
- It has no backend, no server, no database
- It does not use any CSS framework (no Bootstrap, no Tailwind)
- The "AI Assistant" section is NOT real AI — it is a visual placeholder
- The Contact form does NOT send emails — it is frontend-only

## 02 Project Structure

```

MY PORTFOLIO/
├─ index.html      ← Main entry file (20,344 bytes, 624 lines)
├─ 01_mp.html      ← Development source file (20,344 bytes – IDENTICAL to index.html)
├─ style.css        ← All CSS styles (31,578 bytes, 1,582 lines)
├─ script.js        ← All JavaScript (16,876 bytes, 537 lines)
├─ .gitignore       ← Git ignore rules (193 bytes, 19 lines)
├─ README.md        ← Project documentation (6,160 bytes, 130 lines)
├─ assets/
│   ├─ wallpaper.png ← ACTIVE wallpaper (glowing sphere, purple-dark, 416,742 bytes)
│   ├─ wallpaper.jpg ← OLD wallpaper (mountain lake – NOT used, orphaned file)
│   ├─ aditya.jpg    ← Profile photo for About Me window (165,254 bytes)
│   ├─ apple.png     ← Apple logo for menu bar (595,354 bytes)
│   ├─ battery.png   ← Battery icon for menu bar (349,326 bytes)
│   ├─ folder.png    ← Folder icon for all desktop shortcuts (707,025 bytes)
│   ├─ search.png    ← Search icon for menu bar (461,140 bytes)
│   └─ dock/
│       ├─ finder.png ← Dock icon: Finder (617,496 bytes)
│       ├─ safari.png ← Dock icon: Safari (929,055 bytes)
│       ├─ vscode.png ← Dock icon: VS Code (622,185 bytes)
│       ├─ terminal.png ← Dock icon: Terminal (607,509 bytes)
│       ├─ github.png ← Dock icon: GitHub (576,851 bytes)
│       ├─ notes.png  ← Dock icon: Notes (22,007 bytes – smaller than others)
│       ├─ settings.png ← Dock icon: Settings (1,043,678 bytes)
│       └─ trash.png  ← Dock icon: Trash (764,907 bytes)

```

**Total file count:** 22 files (6 root files + 16 assets)

### Notable observations:

- `wallpaper.jpg` still exists on disk but is **not referenced anywhere in the CSS or HTML**. It is an orphaned/unused file left over from an earlier version.
- `notes.png` is only 22,007 bytes, far smaller than other dock icons (which range from 576 KB to 1 MB). This is because Notes came from an old icon strip crop, while the others are full 1024×1024 high-resolution PNGs.
- There is no `package.json`, `node_modules`, `webpack.config.js`, `.babelrc`, or any build tooling whatsoever.

## 03 Technology Stack

| Technology         | What It Does In This Project                                             |
|--------------------|--------------------------------------------------------------------------|
| HTML5              | Document structure, semantic landmarks, accessible forms                 |
| CSS3               | All visual styling, layout, animations, responsive design, window states |
| Vanilla JavaScript | Window management, dragging, z-index, live clock, keyboard shortcuts     |

**Zero external dependencies.** No CDN links, no npm packages, no `<script src="https://...">` tags for libraries.

The font stack is system-native:

```
font-family: -apple-system, BlinkMacSystemFont, "SF Pro Display", "SF Pro Text",  
            "Helvetica Neue", Arial, sans-serif;
```

This means the browser uses **whatever system font the device already has** — no font downloads required.

## 04 HTML File Explained

**File:** `index.html` (and identical `01_mp.html`) — 624 lines

The HTML file is structured using HTML5 semantic landmarks:

### 4.1 `<head>` Section (lines 1–16)

- `<meta charset="UTF-8">` — Tells browsers to use UTF-8 text encoding (handles all characters)
- `<meta name="viewport" ...>` — Makes the site work on mobile phones (prevents zooming out)
- `<meta name="description" ...>` — SEO description that search engines read
- `<title>Aditya Goswami | Portfolio</title>` — Browser tab title
- Favicon: A 🖥️ emoji encoded directly as an SVG data URI (no external favicon file needed)
- `<link rel="stylesheet" href="style.css">` — Loads the CSS file

### 4.2 `<header>` — Menu Bar (lines 18–60)

The top menu bar contains two halves:

**Left side ( `<nav>` ):**

- Apple logo ( `assets/apple.png` ) — 15×15px, purely decorative
- "My portfolio" bold title text
- Five menu items: File, Edit, View, Go, Window — all link to `#` anchors (placeholder, non-functional)

**Right side ( `<div role="status">` ):**

- Search icon ( `assets/search.png` ) — 14×14px
- Wi-Fi icon — **inline SVG** (not an image file)
- Battery icon ( `assets/battery.png` ) — 13×13px
- `<span id="menu-date">` — Date text, updated every second by JavaScript
- `<span id="clock">` — Clock time, updated every second by JavaScript

### 4.3 `<main>` — Desktop Content (lines 61–558)

**Section 1 — Welcome ( `id="welcome"` ):** (lines 63–70)

- The `<h1>` heading: "Welcome to My Portfolio"
- Two `<p>` tags: subtitle text about being an Aspiring Full-Stack Developer
- Note: The two `<p>` tags are outside the `<section>` tag — this is technically non-ideal HTML structure

## Section 2 — Desktop Shortcuts ( `id="desktop"` ): (lines 72–158)

- Contains a `<ul>` with 10 `<li>` items, each containing a `<button>` element
- Each button has `data-window="..."` attribute (e.g., `data-window="about"` )
- Each button has a folder image ( `assets/folder.png` ) and a text label
- These buttons are **fully functional** — clicking them opens portfolio windows

The 10 shortcuts and their `data-window` values: | Label | data-window | |---|---| | About Me | about | | Resume | resume | | Projects | projects | | Skills | skills | | Experience | experience | | Education | education | | Certificates | certificates | | Contact Me | contact | | My Links | links | | AI Assistant | ai |

**Sections 3–12 — Portfolio Windows (lines 160–557):** Each portfolio window is a `<section>` element with:

- `id` matching the `data-window` value (e.g., `id="about"` )
- `role="region"` for accessibility
- `aria-labelledby="title-..."` linking to the window's `<h2>`
- `tabindex="-1"` so JavaScript can programmatically focus it
- A `<div class="window-header">` containing the `<h2>` title

Window order in the DOM (this matters for CSS `nth-of-type`): | DOM position | ID | Content | |---|---|---| | 3rd section | `about` | About Me (two-column: photo + text) | | 4th section | `projects` | Projects (4 articles) | | 5th section | `skills` | Skills (4 categories as cards) | | 6th section | `experience` | Experience (text sections) | | 7th section | `education` | Education (2 articles for B.Tech + HSC) | | 8th section | `certificates` | Certificates (2 placeholder entries) | | 9th section | `resume` | Resume (download link placeholder) | | 10th section | `contact` | Contact (email + form) | | 11th section | `links` | My Links (3 placeholder links) | | 12th section | `ai` | AI Assistant (In Progress placeholder) |

## 4.4 `<footer>` — Dock (lines 559–620)

- Contains 8 `<li>` items, each with a `<div class="dock-item">` element
- Each dock item has a `<img class="dock-icon">` and a `<span class="dock-tooltip">`
- **The dock items are `<div>` elements, NOT `<button>` elements**
- They have NO `data-window` attributes
- They have NO click event listeners
- They are **purely decorative/visual**

The 8 dock icons: | Icon | File | |---|---| | Finder | `assets/dock/finder.png` | | Safari | `assets/dock/safari.png` | | VS Code | `assets/dock/vscode.png` | | Terminal | `assets/dock/terminal.png` | | GitHub | `assets/dock/github.png` | | Notes | `assets/dock/notes.png` | | Settings | `assets/dock/settings.png` | | Trash | `assets/dock/trash.png` (separated by a vertical line) |

## 4.5 `<script src="script.js">` (line 621)

---

The script tag is at the **bottom of the body** — this is correct because it ensures all HTML elements exist before JavaScript tries to find them.

## 05 CSS File Explained

**File:** `style.css` — 1,582 lines, organized into 25 named sections

### Section-by-Section Summary

| CSS Section                 | Lines   | What it Styles                                                             |
|-----------------------------|---------|----------------------------------------------------------------------------|
| 1. Basic Reset              | 10–53   | <code>* { box-sizing: border-box }</code> , body font/background/wallpaper |
| Accessibility Utilities     | 59–74   | <code>.visually-hidden</code> , <code>:focus-visible</code> outline        |
| 2. Header (Menu Bar)        | 76–191  | Fixed top bar, frosted glass, nav, icons, clock                            |
| 3. Desktop Area             | 193–207 | <code>&lt;main&gt;</code> layout base                                      |
| 4. Welcome Area             | 209–245 | H1 styling, white text, subtitle paragraphs                                |
| 5. Desktop Shortcuts        | 247–356 | Left-column folder grid, hover effects                                     |
| 6. Portfolio Windows (base) | 358–430 | Window shape, background, shadow, pseudo traffic lights                    |
| 7. Window Typography        | 432–483 | H2/H3/H4/p/li text styles inside windows                                   |
| 8. About Me                 | 485–543 | Two-column grid layout, photo, badge                                       |
| 9. Projects                 | 545–616 | Article cards, GitHub/Demo link buttons                                    |
| 10. Skills                  | 618–666 | Grid of category cards with pill tags                                      |
| 11. Experience              | 668–679 | Section heading with underline                                             |
| 12. Education               | 681–712 | Card articles for each degree                                              |
| 13. Certificates            | 714–746 | List cards for certificate entries                                         |
| 14. Resume                  | 748–790 | Dashed preview box, blue download button                                   |
| 15. Contact (links)         | 792–816 | "Find Me Online" button links                                              |
| 16. Contact Form            | 818–891 | Form layout, inputs, submit button                                         |
| 17. My Links                | 893–934 | List of social/profile links with hover slide                              |
| 18. AI Assistant            | 936–953 | Gray badge styling for the "In Progress" label                             |

| CSS Section               | Lines     | What it Styles                                                |
|---------------------------|-----------|---------------------------------------------------------------|
| 19. Mac Dock              | 955–1086  | Footer frosted glass, dock icons, tooltips, separator         |
| 20. Scrollbar             | 1088–1108 | Custom webkit scrollbar styling                               |
| 21. Tablet (≤850px)       | 1110–1180 | Responsive adjustments for tablets                            |
| 22. Mobile (≤600px)       | 1182–1333 | Mobile layout, smaller icons, dock scroll                     |
| 23. Small Mobile (≤400px) | 1335–1403 | Extra-small screen adjustments                                |
| 24. JS Window States      | 1405–1563 | <b>Critical:</b> hide/show logic, traffic lights, drag header |
| 25. Reduced Motion        | 1569–1582 | <code>prefers-reduced-motion</code> disables all animations   |

## The Most Important CSS Rules

### The wallpaper (line 32):

```
background: url("assets/wallpaper.png") no-repeat center center fixed;
background-color: #0d0a1a;
```

Uses `wallpaper.png` (the glowing sphere). The `background-color` is the fallback color shown while the image loads.

### The menu bar (line 76–106):

```
header {
  position: fixed;
  top: 0; left: 0;
  width: 100%; height: 28px;
  background: rgba(26, 26, 32, 0.6);
  backdrop-filter: blur(25px) saturate(180%);
  z-index: 1000;
}
```

`position: fixed` keeps it always visible at the top. `backdrop-filter: blur(...)` creates the frosted glass effect. `z-index: 1000` ensures it stays above all windows.

### Desktop shortcuts layout (line 256–287):

```

main > section:nth-of-type(2) {
  position: fixed;
  top: 42px;
  left: 16px;
  z-index: 20;
}
main > section:nth-of-type(2) ul {
  display: flex;
  flex-direction: column;
  flex-wrap: wrap;
  max-height: calc(100vh - 120px);
  gap: 10px 14px;
}

```

The shortcuts section is `position: fixed` (always visible, not scrolled away). `flex-direction: column` with `flex-wrap: wrap` creates the vertical column that wraps into a second column if there are too many icons.

#### Default window hide rule (line 1410–1412):

```

main > section:nth-of-type(n + 3) {
  display: none;
}

```

This hides ALL portfolio windows by default. `nth-of-type(n + 3)` means "the 3rd section and every section after it". This targets all portfolio windows while leaving the Welcome section (1st) and Desktop Shortcuts section (2nd) visible.

#### Window open rule (line 1415–1431):

```

main > section[id].window-open,
main > section:nth-of-type(n + 3).window-open {
  display: block;
  position: fixed;
  top: 80px;
  left: calc(50% - 350px);
  width: 700px;
  max-width: calc(100vw - 32px);
  max-height: calc(100vh - 140px);
  overflow-y: auto;
  z-index: 100;
}

```

When JavaScript adds the class `window-open` to a section, this rule makes it appear as a fixed-position floating window, centered on screen.

#### Dock styles (line 994–1085):

```
.dock-item {  
  width: 64px; height: 64px;  
  cursor: default; /* NOT pointer – confirms non-clickable intent */  
}  
.dock-item:hover {  
  transform: scale(1.18) translateY(-8px); /* macOS bounce effect */  
}
```

## 06 JavaScript File Explained

**File:** `script.js` — 537 lines, organized into 7 numbered sections

### Section 1: Global State (lines 1–17)

```
let highestZIndex = 100; // tracks the topmost window z-index
const desktop = document.querySelector("#desktop");
const portfolioWindows = document.querySelectorAll(
  "main > section[id]:not(#welcome):not(#desktop)"
);
let dragTarget = null; // which window is currently being dragged
let startX, startY, startLeft, startTop; // drag start coordinates
```

`portfolioWindows` is a `NodeList` of all 10 portfolio sections (about, projects, skills, etc.) — it excludes `#welcome` and `#desktop`.

### Section 2: `bringToFront(windowElement)` (lines 22–33)

- Increments `highestZIndex` by 1
- Applies it to the window as inline `style.zIndex`
- Removes `active-window` class from all windows
- Adds `active-window` to the clicked window

This is the z-index stacking system. Every time you click a window or open one, it gets a new, higher z-index so it appears on top of all others.

### Section 3: `activateTopRemainingWindow()` (lines 39–62)

Called when a window is closed or minimized. Scans all open (not minimized) windows, finds the one with the highest z-index, and gives it the `active-window` class.

### Section 4: `clampWindowToViewport(windowElement)` (lines 67–100)

After dragging or resizing, this checks if a window has gone outside the visible screen area and snaps it back within bounds. On mobile ( $\leq 600\text{px}$ ), if the window hasn't been dragged, it clears inline styles to let CSS media queries control the position.

## Section 5: `toggleMaximize(windowElement)` (lines 105–160)

- If already maximized: removes `window-maximized` class, restores saved `left/top/width/height` from `dataset` attributes
- If not maximized: saves current position/size into `dataset`, clears inline styles, adds `window-maximized` class

## Section 6: Desktop Click Listener (lines 163–201)

```
if (desktop) {
  desktop.addEventListener("click", function (event) {
    const trigger = event.target.closest("[data-window]");
    if (!trigger) return;
    // ...opens the window...
  });
}
```

**Event delegation pattern:** One click listener on the whole `#desktop` section.

`event.target.closest("[data-window]")` walks up the DOM from the clicked element to find the button with a `data-window` attribute. This works whether you click the folder image or the text label inside the button.

Opening sequence:

1. Remove `window-minimized` class (in case it was minimized)
2. Add `window-open` class (CSS rule makes it visible)
3. Call `clampWindowToViewport` (ensure it fits on screen)
4. Call `bringToFront` (give it highest z-index, set as active)
5. Store `data-opened-by` for focus return
6. Focus the close button (keyboard accessibility)

## Section 7: Window Controls Setup (lines 206–394)

Loops through every portfolio window and:

1. **Creates traffic-light buttons** dynamically (not in HTML — they're added by JS)
  - Red ( `.window-close` ): closes the window
  - Yellow ( `.window-minimize` ): minimizes (hides) the window
  - Green ( `.window-maximize` ): toggles maximize/restore
2. **Mousedown listener:** clicking anywhere in a window brings it to front
3. **Drag mousedown:** if clicking the top 38px of a window or `.window-header`, starts dragging

4. **Touch start**: same as mouse drag but for touchscreen devices

## Section 8: Global Drag Handlers (lines 399–455)

---

- **mousemove**: if dragging, recalculate window position within viewport bounds
- **touchmove**: same for touch
- **stopDragging()**: resets **dragTarget** to null on mouseup/touchend/blur

## Section 9: Viewport Resize Handler (line 458–467)

---

When the browser window is resized, re-clamps all open windows to stay within the new viewport bounds.

## Section 10: Keyboard Shortcuts (lines 472–483)

---

```
if (event.key === "Escape") {
  // clicks the close button on the active window
}
```

Pressing **Escape** closes whatever window currently has the **active-window** class.

## Section 11: Contact Form (lines 488–493)

---

```
contactForm.addEventListener("submit", function (event) {
  event.preventDefault();
});
```

**event.preventDefault()** stops the browser from actually submitting the form. **There is no email service, no API call, no backend. The form literally does nothing when submitted.**

## Section 12: Live Clock & Date (lines 498–519)

---

```
function updateMenuBarClock() {
  // updates #clock with current time (12-hour AM/PM format)
  // updates #menu-date with "Thu, Oct 2" format
}
updateMenuBarClock(); // runs immediately on page load
setInterval(updateMenuBarClock, 1000); // then every 1 second
```

## Section 13: Hash Link Prevention (lines 524–536)

---

Any `<a href="#">` click is intercepted by `event.preventDefault()` to stop the page from jumping to the top or changing the URL hash. This affects placeholder links in Projects, Resume, My Links, Certificates, and the menu bar items.

## 07 The CSS Specificity Bug (History)

At one point during development, the portfolio windows stopped opening when clicked. This was caused by a **CSS specificity conflict**.

### What CSS Specificity Is

Every CSS selector has a "weight" calculated as (ID selectors, Class/Attribute selectors, Element selectors). A higher-weight rule wins over a lower-weight rule, **regardless of order** in the file.

### The Bug

An older version of the code had this hide rule:

```
/* OLD - BUGGY */
main > section[id]:not(#welcome):not(#desktop) {
  display: none;
}
```

Specificity breakdown:

- **section** = 0,0,1 (element)
- **[id]** = 0,1,0 (attribute selector)
- **:not(#welcome)** = 0,1,0 from **#welcome** ID inside **:not()**
- **:not(#desktop)** = 0,1,0 from **#desktop** ID inside **:not()**
- **Total: 0,3,1**

The override rule:

```
main > section[id].window-open {
  display: block;
}
```

Specificity: **section** (0,0,1) + **[id]** (0,1,0) + **.window-open** (0,1,0) = **0,2,1**

Since **0,3,1 > 0,2,1**, the **display: none** rule **always won**, even after JavaScript added **window-open**. The windows remained invisible no matter what.

## The Fix

---

The hide rule was changed to:

```
/* CURRENT — CORRECT */  
main > section:nth-of-type(n + 3) {  
  display: none;  
}
```

Specificity: `section` (0,0,1) + `:nth-of-type(...)` (0,1,0) = 0,1,1

Now the override `main > section[id].window-open` at 0,2,1 is stronger, so `.window-open` correctly shows the window.

## 08 Window System Deep-Dive

The window system is built entirely with CSS class toggling from JavaScript — no iframes, no modals, no dialog elements.

### States a Window Can Be In

| CSS Classes Present                                                                   | Visual State                                        |
|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| (no classes)                                                                          | Hidden (default — <code>display: none</code> )      |
| <code>window-open</code> + <code>active-window</code>                                 | Open, on top, focused                               |
| <code>window-open</code> (no <code>active-window</code> )                             | Open, in background (slightly dimmed)               |
| <code>window-open</code> + <code>window-minimized</code>                              | Hidden ( <code>display: none</code> wins)           |
| <code>window-open</code> + <code>window-maximized</code>                              | Fullscreen (fills viewport minus menu bar and dock) |
| <code>window-open</code> + <code>active-window</code> + <code>window-maximized</code> | Fullscreen and focused                              |

### Traffic-Light Buttons

The traffic lights (red/yellow/green dots) are NOT in the HTML file. JavaScript **creates them dynamically** during page load and appends them inside each window section. They are `<button>` elements with classes `.window-close` , `.window-minimize` , `.window-maximize` .

When a window is in its default (closed) state, the CSS `::before` pseudo-element draws decorative gray dots to show what the traffic lights will look like. When JavaScript opens the window, the real buttons replace the `::before` dots.

### Drag System

Dragging only works on:

1. The `.window-header` div (the div containing the `<h2>` title)
2. The top 38px of the window (measured from `getBoundingClientRect()` )

Clicking within 38px of the top but not on a `.window-header` starts a drag. Interactive elements inside a window (buttons, inputs, links, textareas) are excluded from drag initiation to prevent accidentally dragging instead of clicking.

## 09 Desktop Shortcuts vs. Dock — Key Difference

This is one of the most important design decisions to understand:

| Feature                            | Desktop Folder Shortcuts                           | Dock Icons                       |
|------------------------------------|----------------------------------------------------|----------------------------------|
| HTML element type                  | <code>&lt;button&gt;</code>                        | <code>&lt;div&gt;</code>         |
| <code>data-window</code> attribute | ✓ Yes                                              | ✗ No                             |
| Opens portfolio windows            | ✓ Yes                                              | ✗ No                             |
| Has click listener                 | ✓ Yes (via <code>#desktop</code> event delegation) | ✗ No                             |
| CSS cursor                         | <code>cursor: pointer</code>                       | <code>cursor: default</code>     |
| Purpose                            | Functional navigation                              | Visual decoration only           |
| Hover effect                       | Folder lifts + background highlight                | Icon scales up + tooltip appears |

**Why was the dock made non-functional?** The user explicitly requested this. The dock represents the "application environment" aesthetic of macOS. In a real Mac, the dock shows running apps — in this portfolio, the dock is purely visual atmosphere. The actual navigation happens through the desktop folder shortcuts on the left side.

## 10 Contact Form — Is It Real?

**No. The contact form does not send any messages.**

Evidence from the HTML:

```
<form action="#" method="post">
```

`action="#"` means the form would just reload the current page if submitted normally.

Evidence from JavaScript:

```
contactForm.addEventListener("submit", function (event) {  
  event.preventDefault();  
});
```

`event.preventDefault()` stops even the page reload. The form submission is completely intercepted and cancelled.

**What the form CAN do:**

- HTML5 `required` validation (browser shows a tooltip if you leave a required field empty)
- HTML5 `type="email"` validation (browser checks that the email field contains an `@` symbol)
- CSS styling: the blue focus ring when you click an input field

**What the form CANNOT do:**

- Send any email
- Store any data
- Call any API
- Show a success/error message

**To make it functional**, you would need to connect it to an email service like Formspree, EmailJS, or Netlify Forms — or write a backend endpoint. None of that has been done.

The README honestly states:

### ★ REMEMBER

"Contact Form: Frontend validation and default submit prevention are implemented; no backend email transport service is attached."

# 11 AI Assistant — Is It Real?

**No. The AI Assistant is a placeholder.**

The complete HTML content of the AI section (lines 544–557):

```
<section id="ai" role="region" aria-labelledby="title-ai" tabindex="-1">
  <div class="window-header">
    <h2 id="title-ai">AI Assistant</h2>
  </div>
  <p><strong>AI Assistant — In Progress</strong></p>
  <p>This feature is currently under development and will be integrated
    in a future version.</p>
</section>
```

That is **all** there is. Two paragraph tags. No chat interface, no input box, no API call, no WebSocket, no backend, no LLM, no model integration.

The README states:

★ **REMEMBER**

"AI Assistant: Displays an authentic 'AI Assistant — In Progress' window placeholder. No backend or LLM API is integrated."

This section exists purely as a portfolio "coming soon" item to show awareness of AI technologies.

## 12 Asset Inventory

### Active Assets (referenced in HTML or CSS)

| File                     | Size     | Referenced In           | Purpose                             |
|--------------------------|----------|-------------------------|-------------------------------------|
| assets/wallpaper.png     | 416 KB   | style.css line 32       | Desktop background                  |
| assets/aditya.jpg        | 161 KB   | index.html line 174     | Profile photo in About Me           |
| assets/apple.png         | 581 KB   | index.html line 22      | Apple logo in menu bar              |
| assets/battery.png       | 341 KB   | index.html line 53      | Battery icon in menu bar            |
| assets/folder.png        | 690 KB   | index.html lines 77–154 | Desktop shortcut folder icons (×10) |
| assets/search.png        | 450 KB   | index.html line 34      | Search icon in menu bar             |
| assets/dock/finder.png   | 603 KB   | index.html line 565     | Dock icon                           |
| assets/dock/safari.png   | 907 KB   | index.html line 571     | Dock icon                           |
| assets/dock/vscode.png   | 608 KB   | index.html line 577     | Dock icon                           |
| assets/dock/terminal.png | 593 KB   | index.html line 584     | Dock icon                           |
| assets/dock/github.png   | 563 KB   | index.html line 593     | Dock icon                           |
| assets/dock/notes.png    | 21 KB    | index.html line 599     | Dock icon                           |
| assets/dock/settings.png | 1,019 KB | index.html line 606     | Dock icon                           |
| assets/dock/trash.png    | 747 KB   | index.html line 615     | Dock icon                           |

### Orphaned Assets (on disk but NOT referenced anywhere)

| File                     | Size   | Status                                                                            |
|--------------------------|--------|-----------------------------------------------------------------------------------|
| assets/<br>wallpaper.jpg | 277 KB | Orphaned. Old mountain-lake wallpaper, replaced by wallpaper.png. Safe to delete. |

## Missing Files (referenced but not present)

---

None — all referenced assets are confirmed present on disk.

## 13 Responsive Design

The CSS has **4 breakpoints**:

### Default (Desktop — no media query, ~1024px+)

---

- Header: 28px tall, all menu items visible
- Desktop shortcuts: fixed left column, 78px wide items with 52×44px folder icons
- Windows: 700px wide, centered, top: 80px
- Dock: 64×64px icons

### Tablet (max-width: 850px)

---

- Header nav menu items ( `File` , `Edit` , etc.) are **hidden** ( `display: none` )
- Windows: stretch to nearly full viewport width ( `calc(100vw - 32px)` ), top: 55px
- Dock icons: shrink to 52×52px

### Mobile (max-width: 600px)

---

- Header height increases to 30px
- Status icons (Search, Battery, Wi-Fi) are **hidden** to prevent overflow
- Desktop shortcuts: switch from **fixed left column** to **static center-wrap grid** (inline layout)
- Folder icons: 44×38px
- Windows: 10px from each side, top: 38px
- Windows on mobile do NOT get default center positioning from JS (JS checks `isMobile` && `mobileDragged` and skips positioning)
- Dock: horizontal scroll (if icons don't fit), tooltips **hidden** ( `display: none !important` )
- Dock icons: shrink to 44×44px

### Small Mobile (max-width: 400px)

---

- Date in menu bar is hidden
- H1 font size reduces to 24px
- Folder items shrink to 60px wide, icons 38×32px
- Dock icons: 35×35px
- Windows: 12px from each edge, even more compact

# 14 Accessibility Features

| Feature                                | Implementation                                                                                                                                                                           |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic HTML landmarks                | <code>&lt;header role="banner"&gt;</code> , <code>&lt;nav&gt;</code> , <code>&lt;main&gt;</code> , <code>&lt;footer aria-label="Application Dock"&gt;</code>                             |
| Visually hidden text                   | <code>.visually-hidden</code> class on <code>&lt;h2&gt;</code> elements inside Desktop and Dock (for screen readers)                                                                     |
| Focus visible outline                  | Global <code>:focus-visible { outline: 2px solid #007aff }</code> — keyboard focus always visible                                                                                        |
| ARIA labels                            | <code>aria-label</code> , <code>aria-labelledby</code> , <code>aria-haspopup="dialog"</code> , <code>aria-pressed</code> on buttons                                                      |
| Keyboard navigation                    | <code>Escape</code> key closes active window; focus returns to the launcher button                                                                                                       |
| Focus management                       | When a window opens, focus goes to its close button. When closed, focus returns to the originating desktop shortcut button                                                               |
| <code>tabindex="-1"</code> on windows  | Allows JavaScript to call <code>.focus()</code> on the section element programmatically                                                                                                  |
| Form labels                            | All form inputs have associated <code>&lt;label for="..."&gt;</code> elements                                                                                                            |
| Required field indicators              | <code>&lt;span aria-hidden="true"&gt;*</code> <code>&lt;/span&gt;</code> — the asterisk is hidden from screen readers (the <code>required</code> attribute carries the semantic meaning) |
| Image alt text                         | All meaningful images have descriptive <code>alt=""</code> attributes; decorative images use <code>alt=""</code>                                                                         |
| Reduced motion support                 | <code>@media (prefers-reduced-motion: reduce)</code> disables ALL animations and transitions                                                                                             |
| <code>rel="noopener noreferrer"</code> | All <code>target="_blank"</code> links have this security attribute                                                                                                                      |

## 15 Things NOT Implemented

These are features that **do not exist** in the current project. This is intentional scope — the project is complete as a static portfolio.

| Feature                      | Status                 | Why                                                           |
|------------------------------|------------------------|---------------------------------------------------------------|
| Contact form backend         | ✗ Not implemented      | Requires a server or email service (Formspree, EmailJS, etc.) |
| AI chat assistant            | ✗ Not implemented      | Requires a backend + LLM API (OpenAI, Gemini, etc.)           |
| Real resume PDF              | ✗ Not implemented      | Download link goes to # placeholder                           |
| Real GitHub/LinkedIn links   | ✗ Not implemented      | All social links go to # placeholder                          |
| Project demo links           | ✗ Not implemented      | All project links go to # placeholder                         |
| Certificate images/links     | ✗ Not implemented      | Placeholder text only                                         |
| Dark/light mode toggle       | ✗ Not implemented      | Always shows dark desktop                                     |
| React / Vue / Angular        | ✗ Intentionally absent | Plain HTML/CSS/JS by design                                   |
| Build system (Webpack, Vite) | ✗ Intentionally absent | No build needed for static files                              |
| Node.js / npm                | ✗ Intentionally absent | Zero server-side code                                         |
| Git repository               | ✗ Not initialized      | <code>git status</code> returns "fatal: not a git repository" |
| CI/CD pipeline               | ✗ Not set up           | Would be GitHub Actions for auto-deploy                       |
| Analytics                    | ✗ Not implemented      | No Google Analytics, no tracking                              |
| Cookie consent               | ✗ Not needed           | No cookies are set                                            |

## 16 Git & Version Control Status

**Git is NOT initialized in this project.**

Running `git status` in the project folder returns:

```
fatal: not a git repository (or any of the parent directories): .git
```

This means:

- There is **no commit history**
- There is **no .git folder**
- There is **no git log**
- Authorship of changes **cannot be verified** through git

The `.gitignore` file **does exist** (19 lines), suggesting git was planned, but the repository was never initialized with `git init`.

The README contains complete instructions for git initialization and GitHub Pages deployment (Section 93–116 in README.md), but none of those steps have been run yet.

**To deploy this project, you would need to run:**

```
git init
git add .
git commit -m "Initial commit - macOS desktop portfolio"
git remote add origin https://github.com/<username>/<repo>.git
git branch -M main
git push -u origin main
```

Then enable GitHub Pages in the repository settings.

## 17 Why Two HTML Files?

The project has both `index.html` and `01_mp.html`. They are **byte-for-byte identical** (same file size: 20,344 bytes, same content).

| File                    | Purpose                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>index.html</code> | Standard entry point required by GitHub Pages, most web servers, and browsers. When you deploy a folder to a web server, the server automatically serves <code>index.html</code> . |
| <code>01_mp.html</code> | Original development filename. "01_mp" may stand for "01 my portfolio" or similar — the naming convention the developer started with.                                              |

**Which one should you open?** Either one works identically. For deployment and sharing, always use `index.html`.

**Going forward:** If you edit one file, copy the exact content to the other. The project convention is to keep them in perfect sync.

## 18 Dependencies & External Libraries

**Zero external dependencies.** This is verified by checking:

- The `<head>` of `index.html` — only one `<link>` tag, for `style.css`
- The `<body>` of `index.html` — only one `<script>` tag, for `script.js`
- No `package.json` file exists
- No `node_modules` folder exists
- No CDN URLs anywhere in any file

The Wi-Fi icon is an **inline SVG** — it's embedded directly in the HTML as vector paths, not loaded from an external file or icon library.

# 19 Performance Considerations

## What's Fast

---

- **Zero HTTP requests for code** — just HTML + CSS + JS (3 files), no library downloads
- **System fonts** — no web font downloads (Google Fonts, etc.)
- **No JavaScript framework overhead** — no React bundle, no virtual DOM
- The `backdrop-filter: blur(...)` CSS effect is GPU-accelerated in modern browsers

## What Could Be Slow

---

- **Large image files** — the dock icons are very large (settings.png is over 1 MB, safari.png is 929 KB). These are unoptimized PNGs. In a production scenario, you would compress them with tools like `squoosh.app` or `pngquant`.
- `assets/folder.png` at 707 KB is loaded for every 10 desktop shortcut buttons (though browsers cache it after the first load)
- **No lazy loading** — `<img>` tags do not have `loading="lazy"` attribute. All images load immediately when the page opens, even the dock icons that may be off-screen

## What Helps with Performance

---

- Reduced motion: `@media (prefers-reduced-motion: reduce)` prevents animations from running for users who have this accessibility setting enabled
- `background: url(...) no-repeat center center fixed` with `background-size: cover` — the wallpaper is one HTTP request
- `pointer-events: none` on decorative elements ( `folder-img` , `.window-header h2` ) prevents unnecessary event processing

## 20 Security Considerations

| Item                                                                         | Status                                                                                            |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>rel="noopener noreferrer"</code> on <code>target="_blank"</code> links | ✓ <b>Present</b> on all external links                                                            |
| No user data collection                                                      | ✓ The form doesn't send data anywhere                                                             |
| No cookies                                                                   | ✓ No <code>document.cookie</code> , no <code>localStorage</code> , no <code>sessionStorage</code> |
| No <code>eval()</code> or <code>innerHTML</code>                             | ✓ JavaScript only uses DOM methods                                                                |
| No external scripts                                                          | ✓ No CDN scripts that could be compromised                                                        |
| HTTPS                                                                        | ✓ GitHub Pages serves HTTPS automatically                                                         |
| Content Security Policy                                                      | ✗ <b>Not set</b> — no <code>&lt;meta http-equiv="Content-Security-Policy"&gt;</code> header       |
| XSS risk                                                                     | ✓ <b>Minimal</b> — no user input is ever rendered into the DOM                                    |

Overall: this is a static site with no user data, no cookies, and no server. Security risk is minimal.

## 21 Deployment Readiness

### What's Ready

---

- `index.html` exists at the root level (required by GitHub Pages)
- `.gitignore` is prepared
- `README.md` contains full deployment instructions
- All asset paths are **relative** ( `assets/wallpaper.png` , not absolute paths)
- CSS uses relative `url("assets/...")` paths — these work correctly on any domain

### What Needs to Be Done Before Deployment

---

1. **Initialize git:** `git init && git add . && git commit -m "Initial commit"`
2. **Create GitHub repository** and push
3. **Enable GitHub Pages** in repository settings (Settings → Pages → Deploy from branch `main` )
4. **Update placeholder links:** GitHub, LinkedIn, resume, project links, certificate links
5. **Optional — Optimize images:** Compress dock icon PNGs to reduce page load time
6. **Optional — Delete orphaned file:** `assets/wallpaper.jpg` is unused and adds 277 KB unnecessarily
7. **Optional — Update README asset list:** The README still lists `wallpaper.jpg` instead of `wallpaper.png`

## 22 README Quality

The README.md is well-structured with:

- Clear project overview
- Feature list
- Technology stack
- Complete project structure diagram
- Local setup instructions (3 methods: direct open, VS Code Live Server, Python server)
- Full GitHub Pages deployment guide with exact commands
- Intentional scope notes (AI placeholder, form placeholder, link placeholders)
- MIT License mention

**One minor inaccuracy:** The README asset tree still shows `wallpaper.jpg` but the active wallpaper is now `wallpaper.png`. The README also doesn't mention `wallpaper.png` in the assets list.

## 23 Project Evolution Timeline

*Note: Git history is unavailable. This timeline is reconstructed from file metadata, content analysis, and development session records. Exact dates of individual changes cannot be verified from the project files.*

| Phase                   | What Was Built                                                                                                                                                                                                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1</b>          | Basic HTML structure with portfolio sections (About, Resume, Projects, Skills, Experience, Education, Certificates, Contact, Links, AI)                                                                                                                                                   |
| <b>Phase 2</b>          | Initial CSS styling — basic desktop layout, window cards, typography                                                                                                                                                                                                                      |
| <b>Phase 3</b>          | JavaScript window management system — open/close/minimize/maximize/drag                                                                                                                                                                                                                   |
| <b>Phase 4</b>          | Accessibility improvements — semantic HTML, ARIA attributes, keyboard navigation, focus management                                                                                                                                                                                        |
| <b>Phase 5</b>          | Visual polish — animations, reduced-motion support, responsive adjustments                                                                                                                                                                                                                |
| <b>Phase 6</b>          | Final QA — bug fixes, code cleanup, GitHub readiness check                                                                                                                                                                                                                                |
| <b>CSS Bug Fix</b>      | Fixed CSS specificity issue where windows wouldn't open (changed hide rule from <code>section[id]:not(#welcome):not(#desktop)</code> to <code>section:nth-of-type(n+3)</code> )                                                                                                           |
| <b>Visual Redesign</b>  | Replaced mountain-lake wallpaper with glowing sphere; added real PNG icons to menu bar; converted desktop shortcuts from horizontal grid to vertical left column; redesigned About Me with two-column photo layout; replaced text-based dock with real PNG app icons; added dock tooltips |
| <b>Dock Bug Fix</b>     | Changed dock items from <code>&lt;button data-window="..."&gt;</code> to <code>&lt;div class="dock-item"&gt;</code> to make dock purely decorative; removed dock click listener from script.js                                                                                            |
| <b>Wallpaper Update</b> | Changed wallpaper from earlier version to current purple glowing planet design                                                                                                                                                                                                            |
| <b>Current State</b>    | Project is complete and deployment-ready (pending git init and real link updates)                                                                                                                                                                                                         |

## 24 Current Status Table

| Feature                             | Status        | Notes                                                             |
|-------------------------------------|---------------|-------------------------------------------------------------------|
| Welcome screen / wallpaper          | ✓ Working     | Purple glowing sphere wallpaper                                   |
| Menu bar                            | ✓ Working     | Apple logo, status icons, live clock/date                         |
| Desktop folder shortcuts (10 total) | ✓ Working     | All 10 open their respective windows                              |
| Dock (8 icons)                      | ✓ Visual only | Decorative — no click behavior                                    |
| Portfolio windows                   | ✓ Working     | All 10 sections open correctly                                    |
| Window close (red button)           | ✓ Working     | Closes and returns focus                                          |
| Window minimize (yellow button)     | ✓ Working     | Hides window, activates next window                               |
| Window maximize (green button)      | ✓ Working     | Fills viewport; restores on second click                          |
| Window drag (mouse)                 | ✓ Working     | Header-only; bounded to viewport                                  |
| Window drag (touch)                 | ✓ Working     | Mobile/tablet touch support                                       |
| Multi-window z-index stacking       | ✓ Working     | Click any window to bring to front                                |
| Escape key closes window            | ✓ Working     | Keyboard shortcut                                                 |
| Focus management                    | ✓ Working     | Focus moves to close button on open; returns to launcher on close |
| Live clock (updates every second)   | ✓ Working     | 12-hour AM/PM format                                              |
| Date display                        | ✓ Working     | "Thu Oct 2" format                                                |
| Responsive design (4 breakpoints)   | ✓ Working     | Desktop / Tablet / Mobile / Small Mobile                          |
| Touch drag on mobile                | ✓ Working     | Passive touch events                                              |
| About Me (two-column with photo)    | ✓ Working     | Photo + text, collapses to single column on mobile                |
| Contact form validation             | ✓ Working     | HTML5 required/email validation                                   |

| Feature                             | Status            | Notes                                         |
|-------------------------------------|-------------------|-----------------------------------------------|
| Contact form submission             | ✗ Frontend only   | <code>preventDefault()</code> — no email sent |
| AI Assistant                        | ✗ Placeholder     | "In Progress" text only                       |
| Resume download                     | ✗ Placeholder     | Link goes to <code>#</code>                   |
| Social/project links                | ✗ Placeholder     | All link to <code>#</code>                    |
| Git repository                      | ✗ Not initialized | No <code>.git</code> folder                   |
| GitHub Pages deployment             | ✗ Not done        | Pending git init and push                     |
| Image optimization                  | ! Not done        | Icons are large uncompressed PNGs             |
| Orphaned <code>wallpaper.jpg</code> | ! Unused file     | Still on disk, not referenced                 |

## 25 Common Recruiter Questions & Answers

**Q** What is this project?

**A** It's a personal portfolio website styled to look like a macOS desktop. Each section of my portfolio — About Me, Skills, Projects, etc. — opens as a draggable floating window, just like an application window on a Mac.

**Q** Why did you build it this way instead of a normal website?

**A** I wanted to demonstrate my understanding of CSS positioning, z-index layering, event-driven JavaScript, and DOM manipulation — not just static content. The macOS interface required me to solve real UI engineering problems like viewport clamping, focus management, and multi-window stacking.

**Q** What technologies did you use?

**A** Pure HTML5, CSS3, and Vanilla JavaScript. No frameworks, no build tools, no libraries. Everything is hand-written.

**Q** How does the window dragging work?

**A** I track `mousedown` events on each window's header area, record the starting cursor position and window position, then use `mousemove` on the document to calculate the delta and update the window's `left` and `top` CSS properties. `mouseup` ends the drag. I implemented the same for `touchstart/touchmove/touchend` for mobile devices.

**Q** Is the AI Assistant real?

**A** No — it's clearly labeled "In Progress" as a portfolio placeholder showing awareness of AI as a field. No API, no backend, no LLM.



**Does the contact form work?**



It has full HTML5 validation (required fields, email format checking), but it doesn't send emails because there's no backend connected. To make it functional, I would integrate it with a service like Formspree or EmailJS.

## 26 Interview Technical Q&A

**Q** Explain CSS specificity and give an example from your project.

**A** Specificity determines which CSS rule wins when two rules target the same element. It's calculated as (IDs, Classes/Attributes/Pseudo-classes, Elements). In my project, I had a bug where `main > section[id]:not(#welcome):not(#desktop) { display: none }` had specificity 0,3,1 because each `:not(#id)` adds ID weight. This prevented my `.window-open` class from overriding it. I fixed it by switching to `main > section:nth-of-type(n+3) { display: none }` which has specificity 0,1,1, allowing `.window-open` at 0,2,1 to win.

**Q** What is event delegation and where do you use it?

**A** Event delegation means adding one listener to a parent element instead of individual listeners on every child. In my project, I have one `click` listener on `#desktop` that handles all 10 folder shortcut buttons. When clicked, `event.target.closest("[data-window]")` finds which button was clicked. This is more efficient and also works for dynamically added elements.

**Q** What does `position: fixed` do in your window system?

**A** `position: fixed` removes an element from the normal document flow and positions it relative to the browser viewport. When I open a window (add `.window-open`), the window becomes `position: fixed` with `top: 80px; left: calc(50% - 350px)` — this centers it on screen and keeps it in place while the user scrolls other content.

**Q** How does z-index stacking work in your multi-window system?

**A** I maintain a `highestZIndex` counter (starting at 100). Every time a window is focused or opened, I increment `highestZIndex` and apply it as `windowElement.style.zIndex = highestZIndex`. The last-focused window always gets the highest number, so it appears on top. The `active-window` CSS class also adds a stronger `box-shadow` to visually indicate the front window.

**Q** How does `backdrop-filter: blur()` work?

**A** `backdrop-filter` applies visual effects to the area behind an element. `blur(25px)` creates a frosted-glass effect — content behind the header or dock appears blurred. This requires the element to have some transparency in its background color (like `rgba(26, 26, 32, 0.6)`). It's GPU-accelerated in modern browsers.

**Q** Why is `pointer-events: none` on the folder images?

**A** The click event on a desktop shortcut should target the `<button>` element, not the `<img>` inside it. With `pointer-events: none` on the image, mouse events "pass through" the image to the button beneath it. Without this, `event.target` would be the `<img>` element, and `closest("[data-window]")` would need to walk further up the tree — or we'd need to handle the image separately.

## 27 Code Walkthrough (Top-to-Bottom)

When the browser loads `index.html`, here is the exact sequence of events:

1. **Browser parses HTML** — builds the DOM tree
2. **Browser loads `style.css`** — applies all styles; windows become `display: none` (hidden)
3. **Browser loads and executes `script.js`** (at bottom of `<body>`):
  - `console.log("SCRIPT JS IS LOADED")` — appears in browser DevTools console
  - `portfolioWindows` NodeList is built (10 sections)
  - Loop runs on each portfolio window — traffic-light buttons are created and appended
  - All drag listeners and click handlers are attached
  - `#desktop` gets its delegated click listener
  - `updateMenuBarClock()` runs once immediately — sets initial time and date
  - `setInterval(updateMenuBarClock, 1000)` — schedules clock to update every second
4. **User sees:** wallpaper + menu bar + desktop shortcuts + dock. All portfolio windows are invisible.
5. **User clicks a folder shortcut:**
  - `#desktop` click listener fires
  - `closest("[data-window]")` finds the button → reads `data-window="about"`
  - `document.getElementById("about")` finds the About Me section
  - `classList.add("window-open")` → CSS makes it `display: block; position: fixed`
  - `bringToFront()` → z-index set to 101, `active-window` class added
  - Focus moves to the close button inside the About Me window
6. **User drags the window:** `mousedown` captures start position → `mousemove` updates `style.left/top`  
→ `mouseup` stops
7. **User presses Escape:** `keydown` listener fires → `querySelector(".active-window")` finds the window  
→ `closeBtn.click()` is called → window closes, focus returns to folder shortcut

## 28 Authorship Assessment

### ★ REMEMBER

"I have personally built most of this project myself. Only some parts were implemented or modified with AI assistance."

#### What can be determined from the project files:

Git history is unavailable (no `.git` folder exists). Therefore, **authorship of specific code sections cannot be verified or attributed** from the project files alone.

The `.gitignore` file exists and was deliberately crafted (it mentions `scratch/` — suggesting awareness of development artifacts). The HTML content (portfolio sections: bio text, education details, skills list, contact email `goswamiaditya112@gmail.com`) is clearly personal information that belongs to Aditya Goswami.

The code demonstrates a range of sophistication — from basic HTML structure to multi-window z-index management, touch event handling, and CSS specificity-aware selectors — consistent with a project developed over multiple sessions with iterative refinement.

**Conclusion:** Authorship of specific code cannot be determined without git history. The portfolio content (personal details, education, skills) is verifiably the developer's own.

## 29 Final What-I-Know Checklist

Use this to confirm your understanding before any interview or explanation:

### HTML

- ✓ I know the difference between semantic HTML elements ( `<header>` , `<nav>` , `<main>` , `<footer>` , `<section>` , `<article>` )
- ✓ I know what `role="region"` , `aria-labelledby` , `aria-haspopup` , `aria-pressed` mean
- ✓ I know why `tabindex="-1"` is used on windows (programmatic focus without tab-navigation)
- ✓ I know the difference between `<button>` (functional) and `<div>` (non-interactive) and why the dock uses `<div>`
- ✓ I know that the traffic-light buttons are **not in the HTML** — they are created by JavaScript

### CSS

- ✓ I can explain CSS specificity with the formula (IDs, Classes, Elements)
- ✓ I know what `position: fixed` does and why windows use it
- ✓ I know what `backdrop-filter: blur()` does (frosted glass)
- ✓ I know what `z-index` does and how the stacking system works
- ✓ I know what `nth-of-type(n + 3)` targets (3rd element and beyond)
- ✓ I know why `display: none` hides windows and `.window-open` makes them visible
- ✓ I know the difference between `@media (max-width: 850px)` and `@media (max-width: 600px)`
- ✓ I know what `@media (prefers-reduced-motion: reduce)` does
- ✓ I can explain why the CSS specificity bug occurred and how it was fixed

### JavaScript

- ✓ I know what event delegation is and why I use it on `#desktop`
- ✓ I know what `event.target.closest("[data-window]")` does
- ✓ I know what `classList.add()` , `classList.remove()` , `classList.contains()` do
- ✓ I know how the drag system works (mousedown → mousemove → mouseup)
- ✓ I know what `getBoundingClientRect()` returns
- ✓ I know what `event.preventDefault()` does and why the form uses it
- ✓ I know what `setInterval(fn, 1000)` does for the live clock

- ✓ I know why the script tag is at the bottom of `<body>`

## Architecture

---

- ✓ I know that this is a **zero-dependency static site**
- ✓ I know that `index.html` and `01_mp.html` are identical copies
- ✓ I know that `wallpaper.jpg` is an orphaned unused file
- ✓ I know that `notes.png` is smaller than other dock icons (from a different source)
- ✓ I know the contact form does NOT send emails
- ✓ I know the AI Assistant is NOT real AI
- ✓ I know git is NOT initialized (no `.git` folder)
- ✓ I know what needs to be done to deploy to GitHub Pages

You've reached the end. Now close this and explain the project out loud, no notes.